



HEREDITY: INHERITANCE AND VARIATION OF TRAITS
(Non- Mendelian Inheritance)

LEARNING MODULE

NAME: _____ YEAR AND SECTION : _____

SET THE TARGET

LEARNING OBJECTIVES:

- At the end of this module, you are expected to:
- 1. Describe the location of genes in chromosomes.
 - 2. Explain the different patterns of non- Mendelian Inheritance.

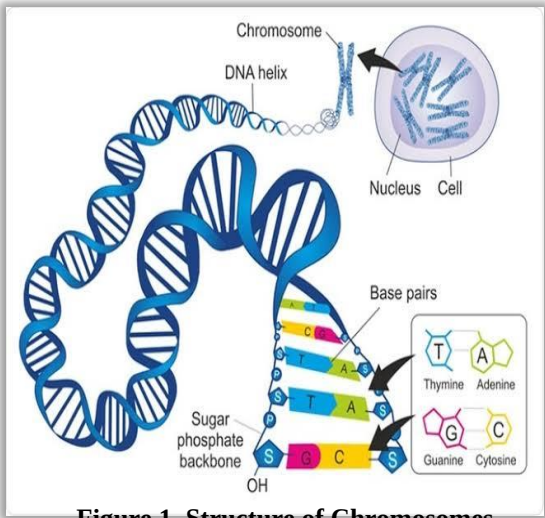


Figure 1. Structure of Chromosomes

Overview:

In your earlier science lessons, you learned that in sexual reproduction, two parents give rise to offspring that have unique combinations of the parents, characteristics. You also learned that each species of living organisms has a characteristic number of chromosomes. The normal human somatic cells have 46 chromosomes (diploid number) while the gametes- sperm and egg have 23 chromosomes from diploid number (2N) to haploid number (N). The process of meiosis maintains the chromosome number of a particular species from one generation to the next.

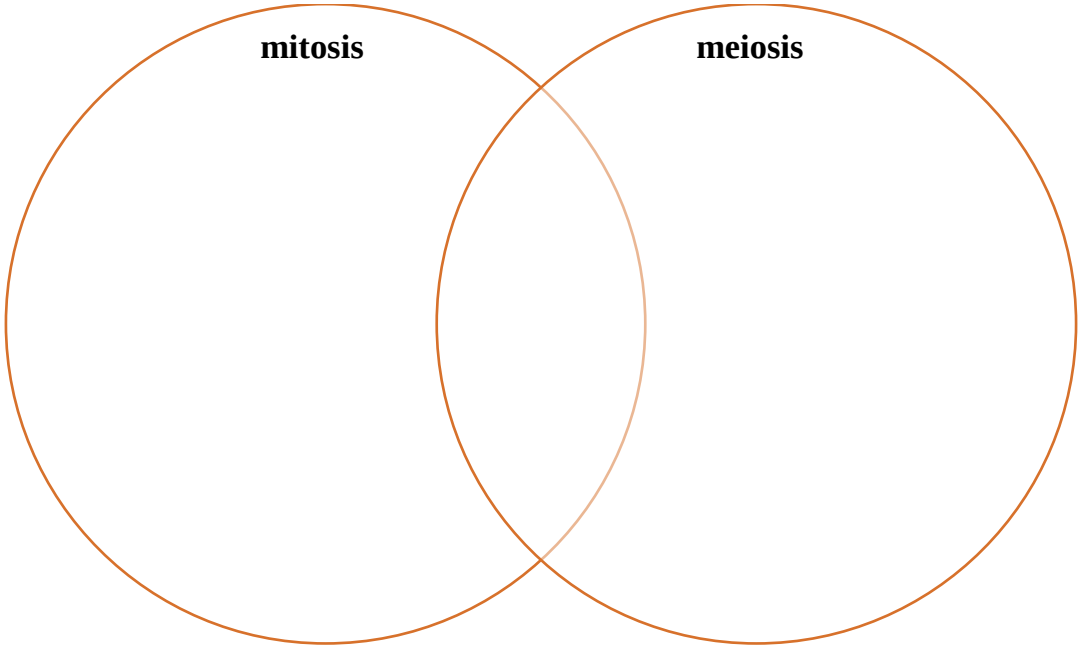
LOOP

1. Recall what you have learned about sexual and asexual reproduction in your early science lessons. Compare and contrast these two methods of reproduction and cite three advantages and disadvantages of each.

Compare and Contrast Chart	
SEXUAL REPRODUCTION	ASEXUAL REPRODUCTION
Similarities	
Differences	

Advantages	
Disadvantages	

2. Give the similarities and differences of mitosis and meiosis using *Venn diagram*.



3. Take a picture of some observable and heritable traits such as earlobe attachment, tongue rolling, cleft chin, dimples, handedness, freckles, naturally curly hair, hairline shape, hand clasping, etc. Classify them according to whom you have inherited the traits. Paste them in the

Father	Mother	Other Relatives

table
belo



LEARN MORE!

THE STRUCTURE OF THE CHROMOSOME

The human body is made up of trillions of cells. Found tightly packed in the nucleus of eukaryotic cells are the chromosomes. Chromosomes are the carriers of genes, the unit determiners of hereditary characteristics. The genes in turn are made up of DNA molecules. The chromosome number varies depending on the species. The table shows the number of chromosomes that compose some species of eukaryotes.

Table 1.0. Chromosome Number of Some Organisms

Species of Organisms	Chromosome Number	Species of Organisms	Chromosome Number
Human	46	Earthworm	36
Gorilla	48	Puffer Fish	42
Donkey	62	Dog	78
Cat	38	Fruit fly	4
Cow	60	Mosquito	6
Goldfish	94	Pea plant	14
Horse	64	Sunflower	34

Each chromosome is made up of two long strands of DNA molecule, the *blueprint of life* or the genetic plan of the organism. The entire DNA in the body cells makes up what is called the human **genome**. Humans have 46 or 23 pairs of chromosomes. The mother contributes 23 chromosomes through the egg cell and the father contributes the other 23 chromosomes through the sperm cell.

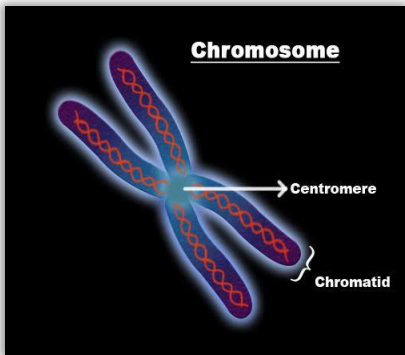


Figure 1.1The centromere holds the pair of chromosomes together

CENTROMERE divides the chromosomes into two “arms”: a long arm region (also called q arm and a short arm region also called p arm region).

CHROMATID- It appears as a single strand or as two identical halves of a single replicated eukaryotic chromosome, the *sister chromatids*, after the DNA is replicated. The two chromatids are joined by the centromere.

GENDER DETERMINATION

Humans have 22 pairs of autosomes, which are the same in both male and female. The last two chromosomes were given letter designations- X and Y. They are the *sex chromosomes*. **Sex chromosomes** determine the gender of the organism. The **Y-chromosome** contains the genes that trigger the development of specific male characteristics. Females have two identical sex

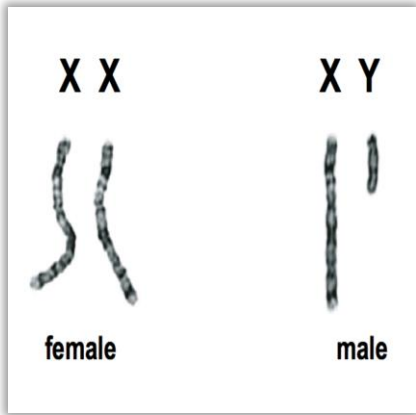
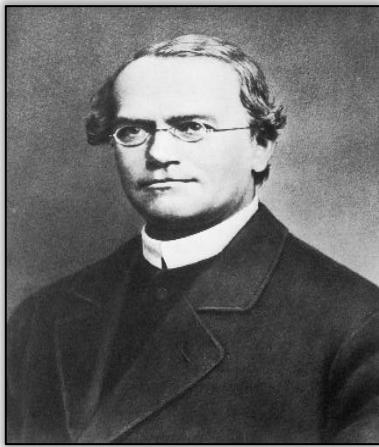


Figure 1.2. The chromosomes of human male and female

chromosomes designated as **XX** and males have two non- identical sex chromosomes designated as **X-** and **Y-** chromosomes.

GENES are the determiners of hereditary characteristics. Sexually-reproducing organisms have different sets of genes for every trait. Genes are portions of the **DNA molecule**, which biologists consider as the fundamental unit of heredity. The number of genes varies from species to species. The specific location of genes along the entire length of the chromosomes is called **locus (Plural: loci)**.



Gregor Johan Mendel
(1822- 1884)

NON- MENDELIAN PATTERN OF INHERITANCE

GREGOR JOHANN MENDEL (1822- 1884) worked with varieties of pea plants to establish the principles and laws that will explain the patterns of trait inheritance among organisms. Mendel's work serve as the foundation of the modern principles of genetics, which earned him the title "*Father of Heredity*". The Mendelian pattern of inheritance were further explored by modern genetics for reason that there were other outcomes of experimentations and inquiries that cannot be answered by the laws of Mendel alone.

HAPLOID is usually represented by $1n$, has half of the amount of normal sets of chromosomes, thus, only one copy of a chromosome set. A human haploid cell has 23 chromosomes. It carries half of the genetic material necessary to form a complete organism.

DIPLOID is opposite of haploid, it is represented by $2n$; a cell that contains two copies of each type of chromosome in its nucleus is formed when the two gametes unite during fertilization. The normal number of chromosomes in a human diploid is 46.

GENETIC REASONING uses probability to predict chances that a certain event may take place. The possible outcomes are the predicted results when two or more traits are combined. The Punnett Square helped the geneticists to predict the possible trait, which the parents can pass on and the number of offspring would inherit such traits when these were combined; one trait from the mother and the other from the father.

PUNNETT SQUARE is a chart used by geneticists to show all possible allelic combinations of gametes in a cross parents with known genotypes. It is a diagram that is used to predict an outcome of a particular cross or breeding experiment. It is named after Reginald C. Punnett, who devised the approach, and is used by biologists to determine the probability of an offspring's having a particular genotype.



Reginald C. Punnett

MONOHYBRID CROSS

Sample problem:

The gene for height in Mendel’s experiment on peas exists in two allelic forms. The allele for tall stature will be designated *T* and for the short stature *t*. The genotype for the homozygous (same allele) tall parent would be *TT*, while that for the homozygous short parent is *tt*.

T	T TT	t Tt
t	Tt	tt

The genotypes in the result are *TT, Tt, Tt, and tt*. The corresponding phenotypes for these genotypes are 3 tall and 1 small.

Another Example:

Geno	Pheno	B	b
BB	Black Brown	BB	Bb
Bb		Bb	bb
bb			

1. Both parents in the above across are: **BLACK**
2. The *Genotype* of both parents in the above cross is.. **Bb**
3. The Genotypic ratio would be: **1:2:1** or **1 BB, 2 Bb, 1 bb**
4. The *Phenotypic* ratio would be: **3:1**

DIHYBRID CROSS

Sample problem:

Set up a punnett square using the following information:

- ✚ Dominant allele for black fur in guinea pigs = B
- ✚ Recessive allele for white fur in guinea pigs= b
- ✚ Dominate allele for rough fur in guinea pigs= R
- ✚ Recessive allele for smooth fur in guinea pigs= r
- ✚ Cross a heterozygous parent BbRr with a heterozygous parent BbRr

	BR	Br	bR	br
BR	BBRR	BBRr	BbRR	BbRr
Br	BBRr	BBrr	BbRr	Bbrr
bR	BbRR	BbRr	bbRR	bbRr
br	BbRr	Bbrr	bbRr	bbrr

1. What is the probability of producing guinea pigs with black, rough fur? **9:16**
Possible Genotype: BBRR, BBRr, BbRR, BbRr
2. What is the probability of producing guinea pigs with black, smooth fur? **3:16**
Possible Genotype: BBrr, Bbrr
3. What is the probability of producing guinea pigs with white, rough fur? **3:16**
Possible Genotype: bbRR, bbRr
4. What is the probability of producing guinea pigs with white , smooth fur? **1:16**
Possible Genotype: bbrr

NAME: _____ YEAR AND SECTION : _____

ACTIVITY 1: PUNNETT SQUARE: MONOHYBRID AND DIHYBRID

OBJECTIVES:

- 1. Solve genetic problems involving different patterns of inheritance.
- 2. Express the genotypes and phenotypes of the offspring formed by the different crosses solved.

MONOHYBRID

- 1. A homozygous yellow pea plant is crossed with a homozygous green pea plant. Yellow is the dominant trait for pea plants

Y= Yellow
Y= Green

y

y

Y

Y

- 1. Both parents in the above across are: _____
 - 2. The *Genotype* of both parents in the above cross is.. _____
 - 3. The *Genotypic* ratio would be: _____
 - 4. The *Phenotypic* ratio would be: _____
- 2. The female dog is heterozygous. The male dog is homozygous recessive. Figure out the phenotypes and genotypes of their possible puppies by using a Punnett Square. Draw your own Punnett Square.

Female Dog: Ff
Male Dog: ff

3. Round seeds are dominant over wrinkled seeds in pea plants. Fill in the Punnett square and determine the expected genotypic and phenotypic ratios from crossing homozygous recessive and heterozygous parents.

DIHYBRID

1. Set up a Punnett square using the following information:
- ✚ Dominant allele for brown eyes = **B**
 - ✚ Recessive allele for blue eyes = **b**
 - ✚ Cross a heterozygous parent with heterozygous parent.

1. How many offspring would be normal coat with brown eyes? _____
Possible Genotype: _____
2. How many offspring would be normal coat with blue eyes? _____
Possible Genotype: _____
3. How many offspring would be back coat with brown eyes? _____
Possible Genotype: _____
4. How many offspring would be black coat with blue eyes? _____
Possible Genotype: _____

2. Set up a punnett square using the following information.

- ✚ Dominate allele for tall plants= **D**
- ✚ Recessive allele for dwarf plants= **d**
- ✚ Dominate allele for purple flowers= **W**
- ✚ Recessive allele for white flowers= **w**
- ✚ Cross a homozygous dominate parent **DDWW** with a homozygous recessive parent **ddww**

- 1. What is the probability of producing tall plants with purple flowers? _____
Possible genotype: _____
- 2. What is the probability of producing dwarf plants with white flowers? _____
Possible genotype: _____
- 3. What is the probability of producing tall plants with white flowers? _____
Possible genotype: _____
- 4. What is the probability of producing dwarf plants with purple flowers? _____
Possible genotype: _____



Concept Micro

Chromosomes are responsible for carrying genetic information in each organism. Gametes are haploid cells produced by meiosis, have 23 chromosomes and diploid cells produced by mitosis have 46 chromosomes.

The **Punnett square** make the passing of trait evident with the use of alleles for each trait and the concept of probability.



TEST YOURSELF!

TEST I. MULTIPLE CHOICE. Write the letter of the best answer on the blank before each number.

_____ 1. When one characteristic is more strongly inherited than another characteristic, it is said to be _____.

- a. monohybrid
- b. dihybrid
- c. dominant
- d. recessive

_____ 2. Phenotypes refers to the _____ of an individual.

- a. actual physical appearance
- b. genetic makeup
- c. recessive alleles
- d. genes

_____ 3. What is the total number of chromosomes in a haploid?

- a. 26
- b. 23
- c. 46
- d. 43

_____ 4. The human sex chromosome for male is

- a. XX
- b. YY
- c. XY
- d. XZ

_____ 5. An individual who is heterozygous for two genes is represented by _____.

- a. Aabb
- b. AAbb
- c. AaBb
- d. AABb

_____ 6. Which one is not an inherited trait?

- a. Birds laying eggs.
- b. Dogs shaking eggs.
- c. Ability of a fish to swim.
- d. Curly or straight tails in pigs.

_____ 7. Which genotype consists of two identical genes for a given character?

- a. homozygous
- b. recessive
- c. dominant
- d. heterozygous

_____ 8. Genes are found.....

- a. in the chromosomes of the cell.
- b. on the cytoplasm the cell.
- c. all over the cell.
- d. In the nucleus of the cell.

_____ 9. Which of the following is an alternate form a gene?

- a. allele
- b. linkage
- c. genotype
- d. phenotype

_____ 10. What type of cell does mitosis create?

- a. Haploid
- b. Sperm cell
- c. Egg cell
- d. Diploid



DEFINITION OF TERMS

GLOSSARY OF TERMS

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DIPLOID is opposite of haploid, it is represented by $2n$; a cell that contains two copies of each type of chromosome in its nucleus is formed when the two gametes unite during fertilization

GENES are the determiners of hereditary characteristics.

GENETIC REASONING uses probability to predict chances that a certain event may take place.

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PUNNETT SQUARE is a diagram that is used to predict an outcome of a particular cross or breeding experiment.



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ELECTRONIC SOURCE

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